

DAY 1 — Linux Foundations: What Linux Is, Filesystem Hierarchy, Distro

Learning Objectives

By the end of today, you will:

- Understand why Linux runs the internet (and why every DevOps/Cloud job needs it)
- Navigate the Linux filesystem confidently
- Know the difference between Ubuntu/Debian and RHEL/CentOS families (this comes up in almost every interview)
- Run your first real commands on a live server or VM

Daily Outcome

You can open a Linux terminal, move around the filesystem, explain what each top-level directory is for, and answer "Why does industry use Linux instead of Windows for servers?" confidently in an interview.

1. Why Linux? (The WHY before the HOW)

Production reality: ~96% of the world's top 1 million servers run Linux. AWS, Google Cloud, Netflix, Amazon, every stock exchange — all Linux. Windows dominates desktops, Linux dominates servers.

Why companies choose Linux for servers:

| Reason | Explanation |

|---|---|

| Free & Open Source | No licensing cost for thousands of servers |

| Stability | Servers run for years without reboot |

| Security | Fewer vulnerabilities exploited at scale, faster patching |

| Automation-friendly | Everything is a file/command — perfect for scripting |

| Resource-light | Runs well even on minimal hardware (important for cloud cost) |

Interview angle: If asked "Why Linux?" — don't recite Wikipedia. Say: *"Because it's stable, scriptable, and cost-effective at scale — a company running 10,000 EC2 instances can't pay per-server Windows licensing, and Linux lets you automate everything via shell/CLI, which is the backbone of DevOps."*

2. Linux Distributions — What You Actually Need to Know

There are hundreds of distros, but **in industry, only two families matter for jobs:**

Family	Distros	Package Manager	Used By
Debian-based	Ubuntu, Debian	<code>apt</code> / <code>apt-get</code>	Startups, AWS default AMIs, most cloud-native shops
RHEL-based	RHEL, CentOS, Rocky Linux, AlmaLinux, Amazon Linux	<code>yum</code> / <code>dnf</code>	Banks, enterprises, government, Fortune 500

Production example: A fintech company (bank) will almost always run **RHEL** because it comes with paid support (Red Hat) and compliance certifications. A startup deploying on AWS will often just use **Ubuntu** or **Amazon Linux 2** because it's free and lightweight.

Interview question you WILL get: *"What's the difference between apt and yum?"*

Both are package managers. `apt` is used on Debian/Ubuntu, `yum` / `dnf` on RHEL/CentOS. Functionally similar (install, remove, update software) but different syntax and different underlying package formats (`.deb` vs `.rpm`).

3. Linux Filesystem Hierarchy Standard (FHS)

Think of it like a company's org chart — every directory has ONE specific job. This is one of the most commonly asked fundamentals in interviews.

```
/
├── bin      → essential user commands
              (ls, cp, cat)
├── sbin     → essential admin commands
              (reboot, iptables)
├── etc      → configuration files (THE
              most important folder for admins)
├── home     → personal directories for
              each user
├── root     → home directory of the root
              user (NOT the / directory!)
├── var      → variable data: logs, mail,
              spool (var/log is huge in real jobs)
├── tmp      → temporary files, cleared on
```

```
reboot
├─ usr      → user-installed programs and
libraries
├─ opt      → optional third-party
software (e.g., custom apps)
├─ boot     → kernel, bootloader files
├─ dev      → device files (disks, USB,
terminals)
├─ proc     → virtual filesystem, live
kernel/process info
├─ mnt/media → mount points for external
drives
└─ lib      → shared libraries needed by
/bin and /sbin
```

Real production example: When an app crashes in production, the FIRST place you check is `/var/log/`. When you need to change how SSH behaves, you edit `/etc/ssh/sshd_config`. When a disk is full, you almost always find the culprit in `/var/log` or `/tmp`.

Analogy: Think of `/` as the root of a company.

- `/etc` = the company's policy manual (all configs)
 - `/var` = the company's daily activity records (logs, that grow every day)
 - `/home` = individual employee desks
 - `/bin`, `/sbin` = the tools every employee/admin needs daily
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4. Commands — Explained Line by Line

```
pwd
```

→ **P**rint **W**orking **D**irectory. Shows exactly where you are right now. Always run this when you're lost.

```
ls -la
```

→ `ls` = list files. `-l` = long format (permissions, owner, size, date). `-a` = show hidden files (files starting with `.`). This is the single most-used command in Linux administration.

```
cd /var/log
```

→ **C**hange **D**irectory. Takes you into `/var/log` where all system logs live.

```
cat /etc/os-release
```

→ Prints file contents. `/etc/os-release` tells you exactly which distro and version you're running — the first command any admin runs on an unfamiliar server.

```
uname -a
```

→ Shows kernel name, hostname, kernel version, architecture — all system info in one line.

```
df -h
```

→ **Disk Free**, human-readable. Shows how much disk space is used/free. First command you run when a server says "disk full."

```
whoami
```

→ Tells you which user you're currently logged in as.

5. Cheat Sheet

Command	Purpose
<code>pwd</code>	Show current directory
<code>ls -la</code>	List all files with details
<code>cd <path></code>	Change directory
<code>cat <file></code>	Show file content
<code>uname -a</code>	Show system info

Command	Purpose
<code>df -h</code>	Show disk usage
<code>whoami</code>	Show current user
<code>hostname</code>	Show machine name
<code>history</code>	Show command history

6. Hands-On Lab

You need: Any Linux environment. Easiest options: WSL2 on Windows, a free-tier AWS EC2 (Ubuntu), or an online terminal like killercode.com / [katacoda](https://katacoda.com) alternative.

Steps:

1. Open your terminal.
2. Run `pwd` → note your current directory.
3. Run `ls -la /` → observe the top-level folders described above.
4. Run `cd /etc && ls` → see the configuration files living here.
5. Run `cat /etc/os-release` → identify your distro.

6. Run `cd /var/log && ls -la` → see real log files.
7. Run `df -h` → check disk usage.
8. Run `cd ~` → return to your home directory. Run `pwd` to confirm.

Expected Output (example on Ubuntu):

```
$ cat /etc/os-release
NAME="Ubuntu"
VERSION="22.04.3 LTS (Jammy Jellyfish)"
```

Common Errors:

| Error | Cause | Fix |

|---|---|---|

| `bash: cd: /etc: Permission denied` | Rare, but check user permissions | Use `sudo` cautiously, or check `ls -ld /etc` |

| `command not found` | Typo or missing package | Double-check spelling; some minimal images lack certain tools |

| Nothing shows in `/var/log` | Some cloud images restrict log read access for non-root | Try `sudo ls -la /var/log` |

Troubleshooting mindset (important for interviews):

When something doesn't work, don't guess — check: *Am I in the right directory? Do I have permission? Does the file/command actually exist?* This 3-question habit is what separates junior from senior engineers.

7. Mini Assignment

1. Take a screenshot (or save output) of `ls -la /` and identify what 5 of those directories are used for, in your own words.
 2. Find and open one log file in `/var/log` (e.g., `/var/log/syslog` or `/var/log/messages`) and describe what kind of information it contains.
-

8. Quiz (10 Questions)

1. What does `pwd` stand for?
2. Which directory holds system-wide configuration files?
3. What's the difference between `/root` and `/`?
4. Name the package manager used by Ubuntu.
5. Name the package manager used by RHEL/CentOS.
6. Where do system logs typically live?
7. What does `df -h` show you?
8. True/False: `/tmp` files survive a reboot.
9. What command tells you which distro you're running?
10. Why do enterprises often prefer RHEL over Ubuntu?

(Answer these yourself first – I'll review them with you before Day 2.)

9. Interview Preparation

Q1: What is Linux and why is it used in servers?

Expected: Open-source OS, kernel + tools, stability/security/automation/cost reasons.

Q2: Explain the Linux directory structure.

Expected: Be able to name and explain at least 6-8 key directories without hesitating.

Q3 (Scenario): A server's disk is full. What's your first step?

Expected: `df -h` to find which partition is full → `du -sh /var/log/*` or similar to find the largest folder → investigate logs/tmp files.

Q4 (Follow-up / cross-question): "What if `/var` is full but `/` is not?"

Expected: Explain that `/var` can be on a separate partition in production, so you check partitions individually.

10. English Communication (30-40 min)

10 Technical Words today:

kernel, distribution, filesystem, directory, permission, package manager, terminal, shell, root, hierarchy

10 Business Words today:

stakeholder, deliverable, escalate, downtime, uptime, SLA, ticket, on-call, rollout, incident

Speaking Task: Record yourself (30-60 seconds)

answering: *"Introduce yourself and explain why you're learning Linux/DevOps."* Focus on clear pronunciation of: kernel, directory, hierarchy.

Writing Task: Write 3-4 sentences describing today's lab in your own words, as if updating your manager in a daily standup. I'll correct every grammar mistake when you share it.

Daily Summary

- Learned why Linux dominates servers
- Learned filesystem hierarchy (`/etc` , `/var` , `/home` , `/bin` , etc.)
- Ran your first navigation and diagnostic commands

- Practiced troubleshooting mindset

Revision Checklist

- ☐ Can explain 8+ top-level directories without looking
- ☐ Can explain apt vs yum
- ☐ Completed the hands-on lab
- ☐ Attempted all 10 quiz questions

Homework

- Finish the Mini Assignment
- Memorize the Cheat Sheet table
- Write your standup-style summary (English task)

Tomorrow Preview

Day 2: Users, Groups, Permissions, and Ownership — the #1 most-tested Linux topic in interviews. You'll learn `chmod`, `chown`, and how real production access control works.

Motivational Note

Every senior engineer once ran `pwd` for the first time too. You don't need to know everything today — you need to be 1% better than yesterday. Let's go. 🚀

When you've completed the lab, quiz, and assignments, share your answers/output here and say "Start Day 2" – I'll review your work first.